Satisfiable sets are consistent

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author corollary labelled cor:consistency-soundness (2523-2526); the proof span 2528-2537 is the author proof environment that immediately follows it.

Editorial note. The source's iftag FOL/PL inline variants are retained as inline alternatives in the packaging; the FOL branch is the counted reading for this unit.

Editorial scope. One original result of the Open Logic Project unit first-order-logic/tableaux, counted once: satisfiability implies consistency. Only the counted corollary and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Consistency, tableau sense. A set Gamma is inconsistent when there are B_1,...,B_n in Gamma and a closed tableau for {True B_1,...,True B_n}; consistent otherwise.. It is the notion whose contrapositive the proof runs (2528-2532).
- Tableau soundness. The unit's soundness theorem: a closed tableau for a finite set of signed formulas is not satisfiable.. It derives unsatisfiability of the finite subset (2532-2536).

Source reference. Original theorem thm-tableau-soundness: sources/olp-first-order-logic-tableaux/pinned-original-source.tex, source lines 2359–2359. The cited result is separately proved within the same frozen unit and delivered as its own corpus card; the reference is contextual packaging, and the counted proof retains its own argumentative steps in full, so the delivery does not depend on the referenced statement for its completeness.

Corollary 0.1. *If Γ is satisfiable, then it is consistent.*

Proof. We prove the contrapositive. Suppose that Γ is not consistent. Then there are $!B_1$, …, $!B_n \in \Gamma$ and a closed !!tableau for $\{\top(!B_1), \ldots, \top(!B_n)\}$. By thm-tableau-soundness, there is no $\mathfrak{A}M$ such that $M \models$ for all $i = 1$, …, n. But then Γ is not satisfiable. □

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